*(نموذج امتحان) *لاب كنترول

1) Draw the block diagram

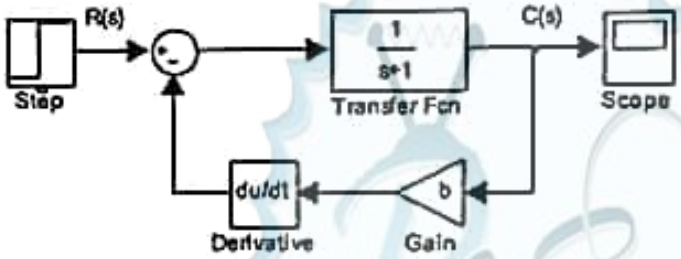
a) $y(t) = 4 + 2t^2 + 3\int y(t)\,dt + 0.5 \sin 3t$

b) $0.4\,\ddot{y}(t) + 5\,\dot{y}(t) + 2z(t) \cdot Y(t) = X(t)$, $X(t) = 3$

$\ddot{z}(t) + 0.2\,\dot{y}(t)\,\dot{z}(t) + 3z(t) = r(t)$, $r(t) = 1$

Scanned by CamScanner

1- Simulate a closed loop system with: G(s)=1/{s+1}; H(s)= bs ,negative feedback and r(t)=unit step,as shown :



2- Plot the output signal on the oscilloscope, and record the results in table 3.4.

Scanned by CamScanner

(4) $y''(t) - 2y(t) - 5y(t) - 2m(t)$

A) solve → state space.

A [ ]

B [ ]

c [ ]

d [ ]

B) solve → Transfer function

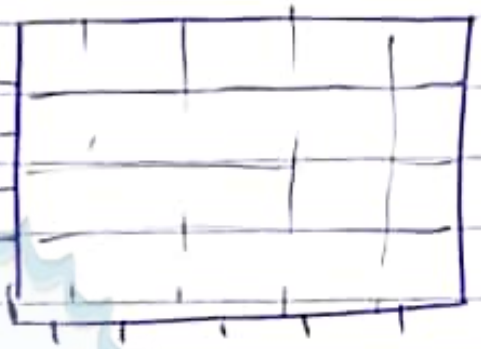
Scanned by CamScanner

مبدكنترول [ملاب]
طبيعة الامتحان ←
عملي و نظري

نظري ← سؤالين

1 رسم شكل الـ fun في حدود كالآتي

1) r(t) = 1, T = 1, a = 1
2) r(t) = 2, T = 1, a = 1
3) r(t) = 1, T = 1, a = 0
4) r(t) = 1, T = 5, a = 1



2 رسم الـ block diagram للمعادلات

a) من المحاضرات الموضوع
Solving Linear differential eqns. by integration

b) من المحاضرات الموضوع
Steady State

العملي ← Time Response of 1's order ← تمثيل

r(t) = 1, T = 0.5, 1, 2, Transfer fun $\frac{1}{Ts+a}$
a = 1

- رسم - جدول يوضح القيم والـ error - الاستنتاج

Scanned by CamScanner

الامتحان كان عبارة عن سؤالين نظري وسؤال عملي

النظري

الاول كان تمثيل اقترانين ببلوكات زي كانه عملي
طبعا الاقترانين تقريبا موجود نفسهم بتلخيص الpce (التجربة الثانية) واحد الي بيجي فيه اقترانين ومندخل المكس فيهم وواحد زي اول حالة بالدوسية وكان من ضمن الاقتران ديريفاتف يعني بدنا ندخل بلوك مشتقة (او ممكن يجيب تكامل)

السؤال الثاني اعطانا اقتران
اول فرع منه انحله عن طريق الستيت سبيس ونطول المصفوفات (بعد ما انطولهم بدنا انعبيهم عالورقة زي ما بنعبيهم عملي مع فواصل وكذا)

الفرع الثاني احل نفس الاقتران بطريقة الترانسفير فنكشن (يعني اطول الترانسفير فنكشن)

العملي

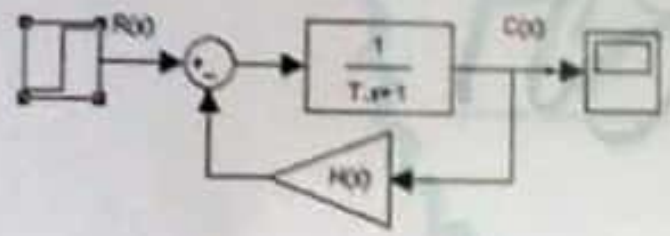
بعطينا اقتران عشان انمثله

وفي 3 حالات كل حالة بتغير فيها الفيد باك

بطلب نعبي القيم لكل كيرف من ال3 كيرفات بالجدول(عند الزمن كذا كم قيمة الاوتبوت..)
وبعدين منرسمهم الثلاث

وبعدين منكتب ملاحظات عن الي صار

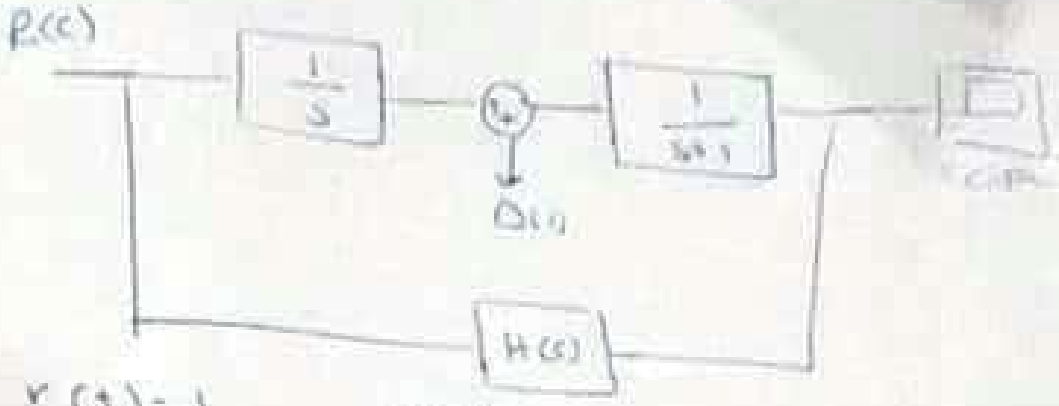
Q1) (12 points) Plot, in scale c(t)



a] r(t)=1; H(s)=0; T=1
b] r(t)=1; H(s)=1; T=1
c] r(t)=2; H(s)=0; T=1
d] r(t)=1; H(s)=1/s; T=1

Scanned by CamScanner

Q: 1



r(t) = 0
d(t) = 0
H(s) = 1

r(t) = 1
d(t) = 0
H(s) = 0

r(t) = 0
d(t) = 1
H(s) = 1

r(t) = 1
d(t) = 1
H(s) = 0

Q: 2

a) $\dddot{y}(t) + 2\ddot{y}(t) = 3u(t) + 4r(t)$

A. [ ]

B [ ]

C. [ ]

D. [ ]

b) Solve as equation using simulation diagram

Scanned by CamScanner

Name ID:

Q2) (8 points) a DC servo motor is given as

$$v(t) = Ri(t) + ldi(t)/dt + \omega(t)$$

$$Jd\omega(t)/dt = i(t) - i_l(t)$$

A) solve as a set of equations using simulation diagram

B) represent in state space form

Scanned by CamScanner

Al-Balqa' Applied Uni...
Faculty Of Engineering Technology
Mechatronics Engineering Dept.
Student ID: 3115018023

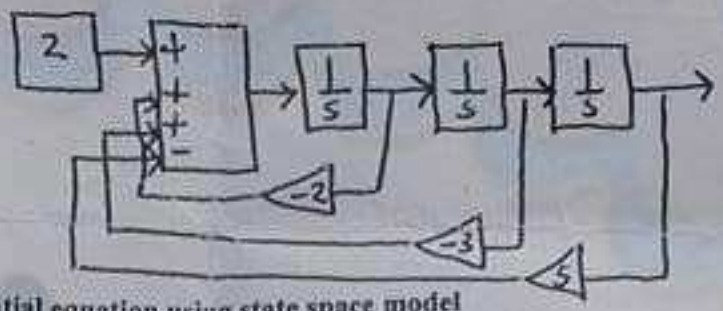
Control Theory Lab
Instructor name:

18

اسم الطالب: اشرف خالد البيشي

mid Exam
20 points

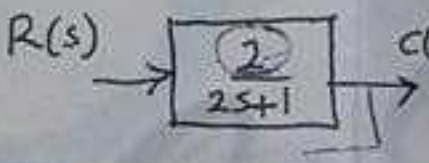
1. The differential equation that can be solved by the following model is
   a. $\dddot{Y}(t) + 2\ddot{y}(t) + 3\dot{y}(t) - 5y(t) = 2$
   b. $\dddot{Y}(t) + 2\ddot{y}(t) + 5\dot{y}(t) + 3y(t) = 2$
   c. $2\dddot{Y}(t) - 2\ddot{y}(t) - 3\dot{y}(t) + 5y(t) = 2$
   (d.) $\dddot{Y}(t) + 2\ddot{y}(t) + 3\dot{y}(t) + 5y(t) = 2$



2. If we represent the following differential equation using state space model
   $\dddot{Y}(t) + 3\ddot{y}(t) + 5\dot{y}(t) + 2y(t) = r(t)$, $r(t) = 2$
   The matrix A =
   a. $\begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & -3 \end{vmatrix}$
   b. $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -5 & -3 \end{vmatrix}$
   c. $\begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 5 & 3 \end{vmatrix}$
   $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & -5 & -2 \end{vmatrix}$
   $\begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -5 & -2 \end{vmatrix}$

3. The transfer function that represent this differentialequation
   $\dddot{Y}(t) + 3\ddot{y}(t) + 5\dot{y}(t) + 2y(t) = r(t)$, $r(t) = 2$
   a. $\frac{2}{S(s^3+3s^2+5s+2)}$
   b. $\frac{1}{S(s^3+3s^2+5s+2)}$
   c. $\frac{1}{(s^3+3s^2+2s+5)}$
   (d.) $\frac{1}{(s^3+3s^2+5s+2)}$
   e. $\frac{2}{(s^3+3s^2+5s+2)}$

***for the following system



4. The time required to reach the steady state value =
   a. 2-3 sec
   b. 4-5 sec
   c. 6-8 sec
   (d.) 8-10 sec

5. If r(t) = 2 , the final value =
   a. 1
   b. 2
   c. 3
   (d.) 4

***if a negative feedback H(s) = 2 was added , answer the following

6. The total transfer function become =
   a. $\frac{2}{2s+2}$
   b. $\frac{4}{2s+1}$
   (c.) $\frac{2}{2s+5}$
   d. $\frac{2}{2s+4}$
   e. $\frac{1}{s+3}$



7. The gain of the system
   a. Increased
   (b.) decreased
   c. not effected

8. The system become
   a. Faster
   b. slower
   (c.) not effected

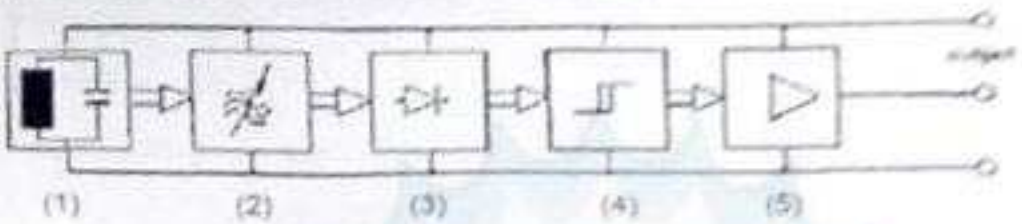
9. The final value of the system when r(t) = 3 is
   a. 1
   b. 3
   c. 12
   (d.) 1.2

*** To represent the transfer function $\frac{s+1}{s^2+2}$ in mat lab Simulink, the numerator matrix = [1 1].

The denominator matrix = [1 0 2 1]

1	2	3	4	5	6	7	8	9
d	e	d	d	d	c	b	c	d

Q2: Giving the block diagram of the inductive proximity switch .



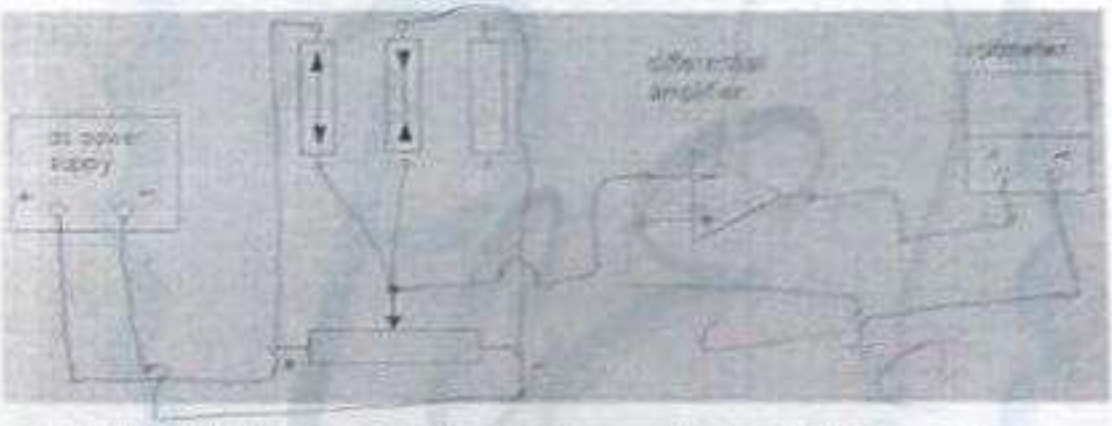
a) write the name of the elements from 1 to 5. (2.5 marks).

1 LC circuit 2 Oscillator 3 Demodulator 4 trigger 5 Amplifier

b) What are the factors that switching distance depends on? (2.5 marks)

1 type of metal (Metal Permeability) 2 Voltage Coil diameter 3 temperature 4 effective surface

Q3: Connect the elements in the following to obtain a half-bridge strain gauge system. (2.5 marks).



Q4: the graph below represents the output of the strain gauges (SGs) experiments,

1) Yaxis (output) represents Voltage in [mV]

2) Xaxis(input) represents displacement in [mm] (distance)

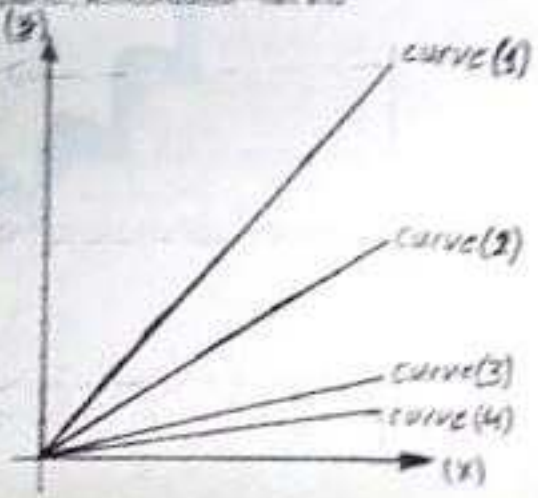
Denote which curves represent the output of SGs (wire-half and quarter, semiconductor –half and quarter)? (1.5 marks).

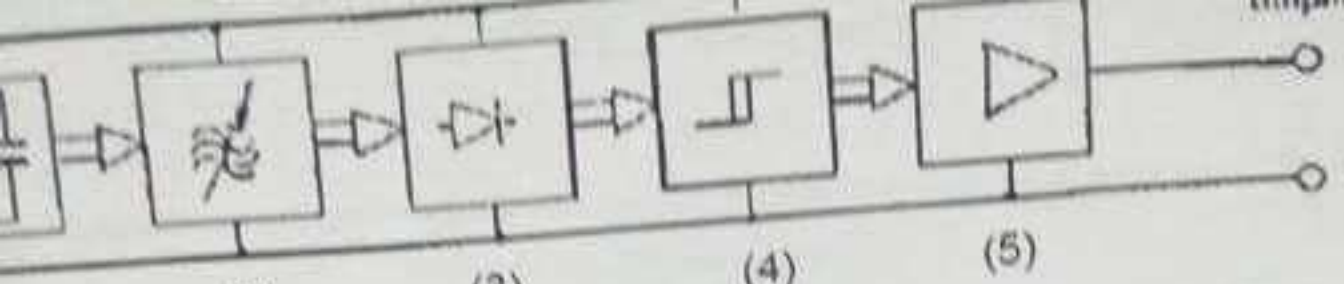
3) Curve 1 represents half Semiconductor

4) Curve 2 represents Quarter Semiconductor

5) Curve 3 represents half wire

6) Curve 4 represents Quarter wire





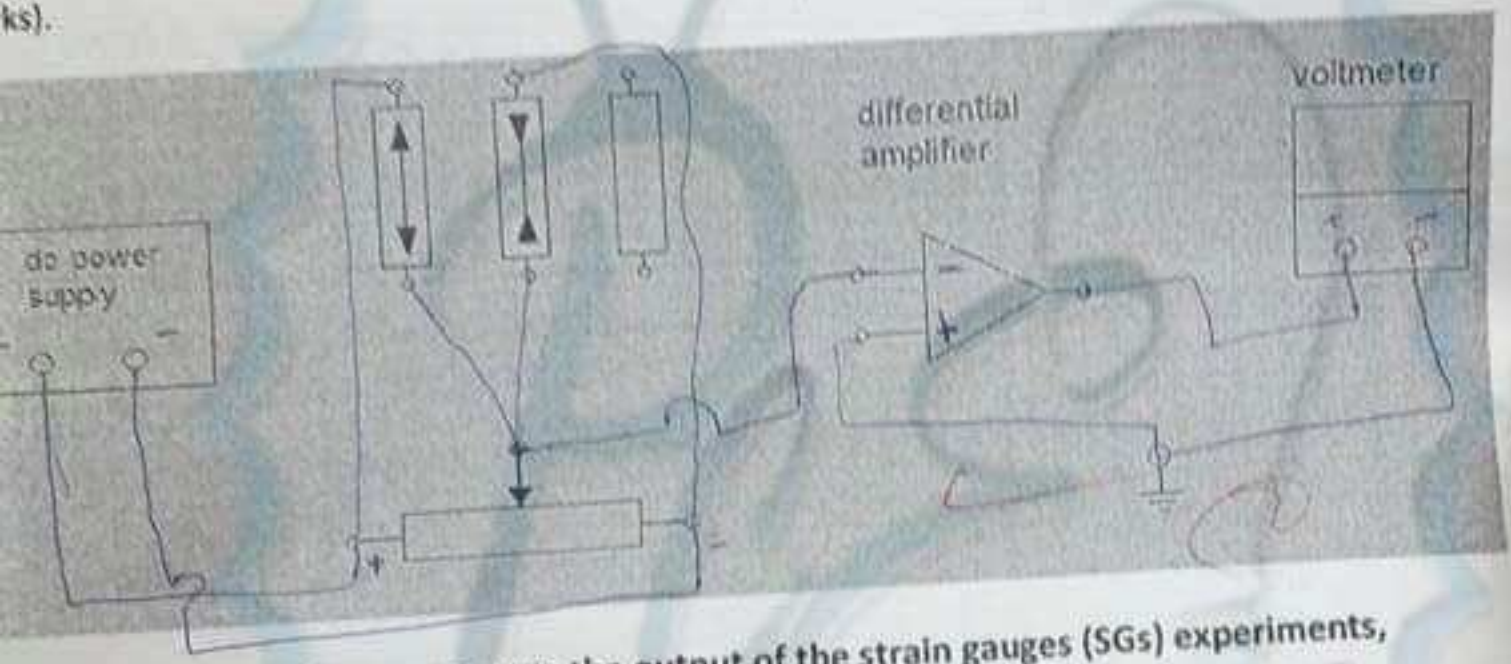
write the name of the elements from 1 to 5. (2.5 marks).

1 c circuit 2 Oscillator 3 Demodulator 4 trigger 5 Amplifier

) What are the factors that switching distance depends on? (2.5 marks).

1 type of metal (Metal, Brass, Al ...) 2 ~~Voltage~~ Area coil diameter 3 temprature 4 objective surface

Connect the elements in the following to obtain a half-bridge strain gauge system. (2.5 ks).



Q4: the graph below represents the output of the strain gauges (SGs) experiments,

1) Yaxis (output) represents Voltage in [mV]

2) Xaxis(input) represents displacment in [mm] (distance)

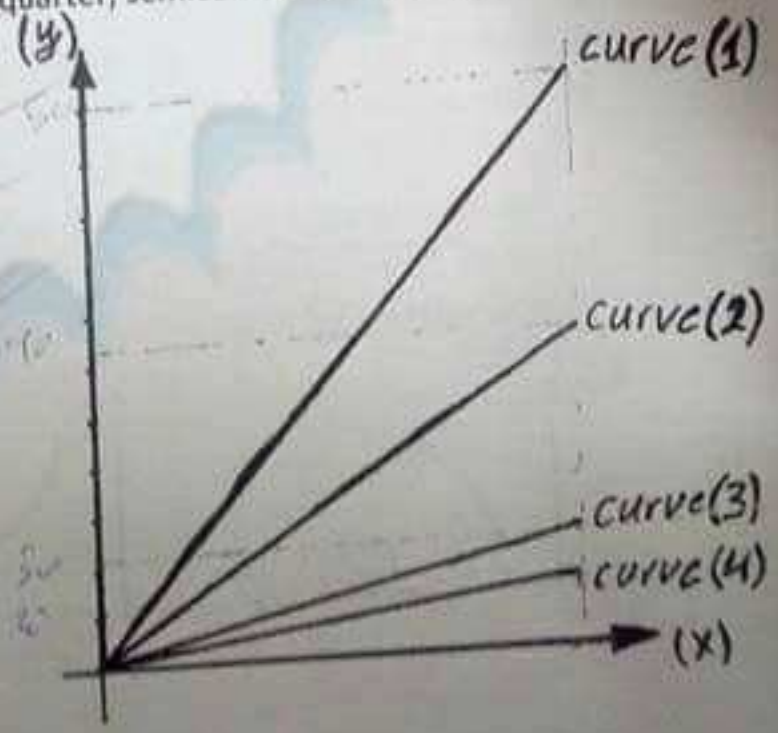
Denote which curves represent the output of SGs (wire-half and quarter, semiconductor –half and quarter)? (1.5 marks).

3) Curve 1 represents half Semiconductor

4) Curve 2 represents quarter Semiconductor

5) Curve 3 represents half wire

6) Curve 4 represents quarter wire



Choose the correct answer using MATLAB.

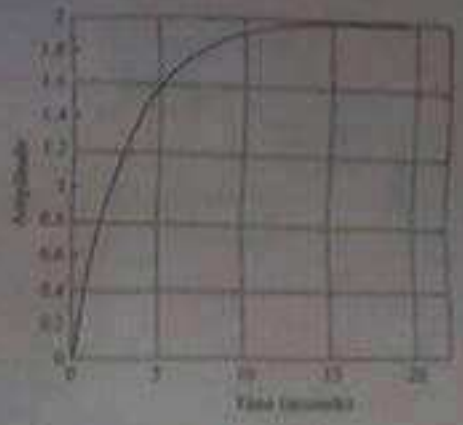
Q1)

Closed-loop transfer function of a unity-feedback system is given by $Y(s)/R(s) = 1/(\tau s + 1)$. Steady-state error to unit-ramp input is

(A) ∞ (B) τ (C) 1 (D) $1/\tau$

Q2)

The plot in Fig. shows the unit step response of a first-order system. Transfer function of the system is



(A) $\frac{1}{5s+1}$ (B) $\frac{2}{15s+1}$

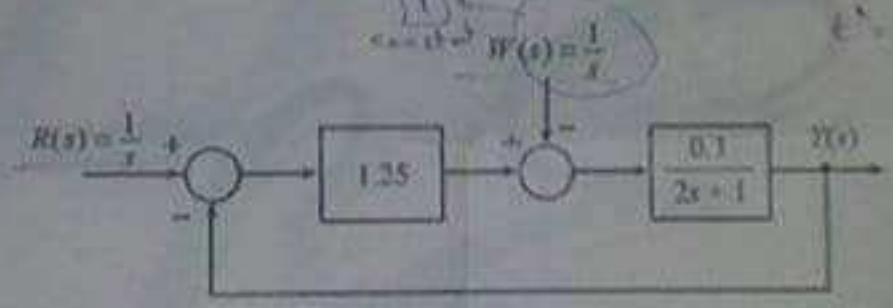
(C) $\frac{2}{3s+1}$ (D) $\frac{1}{15s+1}$

Q3)

The system of Fig. has the steady-state error

(A) $\frac{1}{1.125}$ (B) $\frac{0.1}{1.125}$

(C) $\frac{1.1}{1.125}$ (D) None of the answers in (A), (B), and (C) is correct

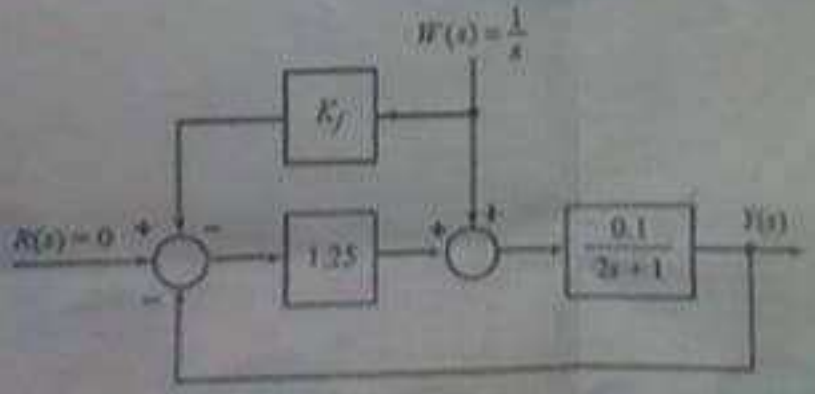


Q4)

Consider the system of Fig. . The steady-state error is zero for

(A) $K_f = 0$ (B) $K_f = \frac{1}{1.25}$

(C) $K_f = 1.25$ (D) None of the answers in (A), (B), and (C) is correct



Q5)
The open-loop transfer function of a unity feedback system is

$$G(s) = K/[s^2(s+5)];\ K>0$$

The system is unstable for

(A) $K > 5$
(B) $K < 5$
(C) $K > 0$
(D) All the answers in (A), (B), and (C) are correct

Q6)
A unity feedback system with forward path transfer function $G(s) = 1/[s^2(s+1)]$ is subjected to an input $r(t) = K_1 + K_2 t + \frac{1}{2}t^2$. The steady-state error of the system is

(A) infinity
(B) 1
(C) zero
(D) None of the answers in (A), (B), and (C) is correct

Q7)
A state variable formulation of a system is given by the equations

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [0]u$$

The transfer function of the system is

(A) $\dfrac{1}{(s+1)(s+3)}$
(B) $\dfrac{1}{s+1}$
(C) $\dfrac{1}{s+3}$
(D) None of the answers in (A), (B), and (C) is correct

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	C	D	B	D	A	D

10/20

B C C B D B B